



SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	18/05/2026
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	18/05/2027

Assigned user to Sign Off	Oraib AlKheetanRojas
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	

Operation Assessed	Operation of Unmanned Aerial Vehicle (Drone) - Outdoors
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Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

This risk assessment covers the outdoor operation of unmanned aerial vehicles (UAVs/drones) for research, testing, data collection, demonstrations, and payload-based activities. Work may include transport to site, pre-flight checks, battery installation and removal, payload mounting, take-off and landing, manual or assisted flight, ground-station operation, data capture, post-flight inspection, and return of equipment to storage. Activities may take place in approved outdoor test areas or controlled field locations under suitable weather, visibility, and site conditions.

Reason for Risk Assessment

To identify, assess, and control health, safety, collision, propeller, battery, fire, equipment-damage, environmental, public-interface, and operational risks associated with outdoor UAV/drone operation during research, testing, demonstration, and data-collection activities.

Overall Current Risk

High Risk



Average Number of Persons Affected

4  Employees

4  Students

Current Rating	Hazard Information	Task Status
Moderate Risk 	Fall of Object from Height The UAV may fall from height due to loss of control, battery failure, GPS/communication failure, software error, pilot error, adverse weather, collision, or mechanical malfunction. A falling UAV or payload may strike people, vehicles, buildings, or equipment, causing injury or damage.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Pre-flight checks are completed before each flight. - UAV, propellers, frame, payload mounts, batteries, and connections are inspected before use. - Flights are carried out only by trained/authorised operators. - Flights are conducted only in approved outdoor areas with suitable space and clear separation from people, vehicles, and buildings. - Take-off and landing areas are controlled and kept clear of non-essential personnel. - Weather conditions are checked before flight; flights are not conducted in unsuitable wind, rain, poor visibility, or unstable conditions. - Battery level and flight time are monitored during operation. - Payloads are securely mounted and checked before take-off. - Emergency landing/return-to-home procedures are available where supported by the UAV.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Other (See: Hazard Details) Privacy invasion. Outdoor UAV operation may unintentionally capture images, video, audio, location data, or other information relating to people, vehicles, buildings, or restricted areas. This may create privacy, confidentiality, reputational, or regulatory concerns if data is captured, stored, shared, or used inappropriately.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- UAV flights are planned in advance and limited to approved research, testing, or demonstration purposes. - Operators avoid unnecessary recording of people, private areas, restricted spaces, or unrelated activities. - Camera/sensor use is limited to the minimum required for the activity. - Data capture areas are selected to reduce privacy impact where possible. - Collected data is stored and handled in accordance with lab, university, and project requirements. - Permissions/approvals are obtained where required before filming or data collection. - Non-essential footage containing identifiable people or unrelated activities is deleted or excluded where appropriate.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	⚠ Fire UAVs commonly use rechargeable lithium-based batteries. Battery damage, over-discharge, crash impact, short circuit, incorrect charging, overheating, or electronic failure may cause smoke, fire, thermal runaway, toxic fumes, or damage to the UAV, nearby equipment, vegetation, or property.	0 0 0
Measures Currently in place to prevent risk of injury		
- Batteries are inspected before use for swelling, damage, leakage, odour, heat, corrosion, or abnormal condition. - Damaged, swollen, leaking, hot, crashed, or suspect batteries are not used. - Batteries are charged only using compatible chargers and correct battery-specific settings. - Battery charging is covered under the rechargeable battery charging RA/SOP. - Battery level and temperature/condition are monitored before and after flight where possible. - UAV is landed as soon as safe if battery warnings, abnormal behaviour, smoke, odour, or overheating are observed. - Combustible materials are kept away from battery charging and preparation areas. - Fire/emergency procedures are followed in the event of smoke, fire, or battery failure.		

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
High Risk 	⚠ Moving Parts UAV propellers rotate at high speed and may cause cuts, lacerations, eye injury, entanglement, or impact injuries. The UAV itself is also a moving object and may collide with people, vehicles, structures, or equipment during take-off, landing, flight, or loss-of-control events.	0 0 0
Measures Currently in place to prevent risk of injury		
- Only trained/authorised operators may operate the UAV. - Pre-flight checks include propellers, motors, guards where fitted, frame, payload, and control systems. - Propellers are not touched while the UAV is powered or armed. - UAV is powered/armed only when positioned in the designated take-off area and ready for flight. - Take-off and landing zones are kept clear of non-essential personnel. - Safe separation is maintained between the UAV and people, vehicles, buildings, and equipment. - Operators maintain visual awareness of the UAV and surrounding area during flight. - Flights are stopped or postponed if people enter the operating area or conditions become unsafe. - Damaged propellers, motors, guards, or frame components are replaced before further use.		

Potential Rating	Additional Controls Required
High Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.